RESEARCH, INNOVATION, INDUSTRY, SOCIETAL AND SDG IMPACT REPORT

1. RESEARCH IMPACT

The project creates a reproducible machine-learning workflow for fake-profile detection. It provides a foundation for comparing tabular classifiers and investigating behavioral intelligence.

2. INNOVATION IMPACT

Innovation comes from combining profile features, behavioral indicators, risk scoring and explainable predictions. The architecture also separates prediction from human review.

3. INDUSTRY IMPACT

Social platforms and digital communities can use risk scores to prioritize moderation queues. Similar approaches can support fraud prevention, spam filtering, impersonation detection and trust-and-safety operations.

4. SOCIETAL IMPACT

Fake accounts can contribute to scams, harassment, impersonation and manipulation. Earlier identification can reduce exposure and help moderators focus on higher-risk cases. Because classification can be imperfect, automated predictions should support rather than replace appropriate human review.

5. SDG 9

The project supports innovation and AI-enabled digital infrastructure.

6. SDG 11

The project contributes to safer and more resilient digital communities.

7. SDG 16

The project supports trustworthy digital environments and can help reduce technology-enabled abuse.

8. ETHICAL CONSIDERATIONS

False positives may affect legitimate users. Training data can contain bias. The model should be audited, monitored and evaluated across relevant user groups and conditions. Data minimization and privacy protection are important.

9. FUTURE RESEARCH

Graph Neural Networks, transformer-based content analysis, multimodal models, continual learning, federated learning, adversarial robustness and real-time stream processing are promising directions.

10. CONCLUSION

The project has both technical and social value because it addresses an important trust-and-safety problem while demonstrating practical AI engineering.